cdmetapopR: A companion R package for the CDMetaPOP simulation software

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Summary

This R package is a companion to CDMetaPOP, an individual-based, spatially explicit population genetics and demography simulation model built on python architecture (Landguth et al. 2017; Day et al. 2023). The package includes four Shiny applications that facilitate the construction of the four fundamental input files required by CDMetaPOP. It also provides a function to launch CDMetaPOP directly from R, and several functions for visualization and analysis of CDMetaPOP output.

Statement of need

This package is designed to facilitate the use of CDMetaPOP by an audience comprised of researchers, including evolutionary biologists, landscape ecologists, geneticists, and epidemiologists. cdmetapopR provides an R-based platform for new users interested in applying CDMetaPOP to their own study systems. By offering user-friendlier tools for first-time implementation, the package lowers the barrier to entry and is expected to promote broader adoption of CDMetaPOP across multiple research disciplines.

Software design

The package provides four Shiny applications that can be used to construct the core input files required by CDMetaPOP (e.g. write_runvars(); write_popvars(); Figure 1). Each Shiny app offers a streamlined approach to input-file creation by guiding users through a series of choices in a graphical user interface, referencing help from the CDMetaPOP manual when needed, and modifying standard default input files based on those selections. Once users have completed their inputs within a Shiny app, they can write the resulting CSV file locally and use it to launch CDMetaPOP.

The launch_cdmetapop() function launches a command prompt (Windows) or terminal (Linux/Mac) that will call Python and supply the five arguments needed to launch CDMetaPOP simulations (Figure 1).

The five arguments are: 1) the filepath to the Python executable, 2) the filepath to the CDMetaPOP.py file, 3) the filepath to the directory where the input files are stored for the run, 4) the name of the RunVars file to run, and 5) the name of the output directory to be stored in the same location as the input directory. This function returns an output directory with the CDMetaPOP output files, which can then be analyzed using the package's built-in functions or other tools.

The built-in functions for analyzing CDMetaPOP outputs are organized into seven major classes: 1) summary_dataframe(), which reads CDMetaPOP output files as data frames for custom plotting or analysis; 2) summary_pop(), which summarizes population demographics and genetics from summary_popAllTime.csv; 3) summary_class(), which reads and summarizes age and size class dynamics from summary_classAllTime.csv; 4) summary_ind(), which computes metric scores at the individual level from 'indxx.csv' output files, including movement, demographic histograms, and age/size structure dynamics; 5)

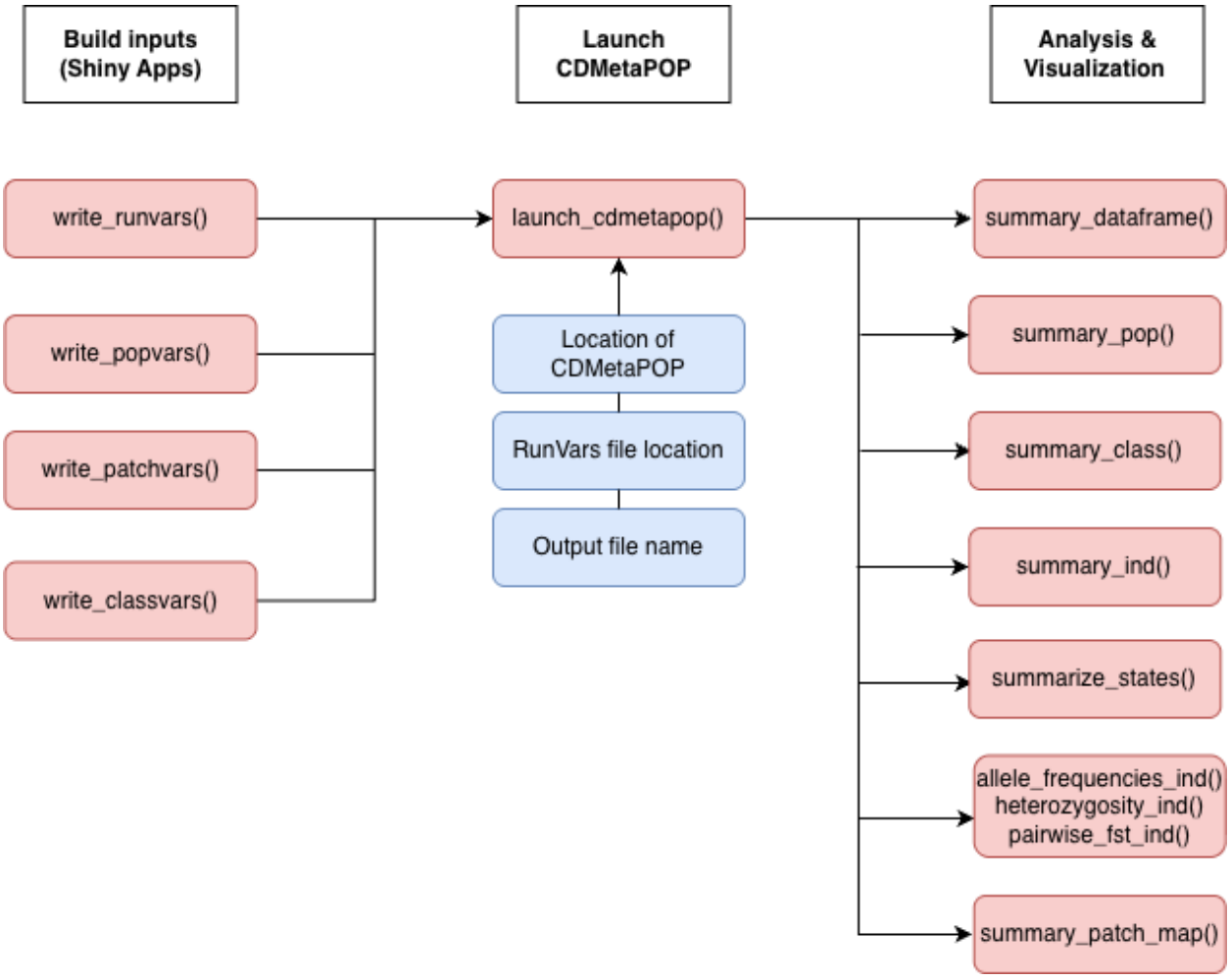
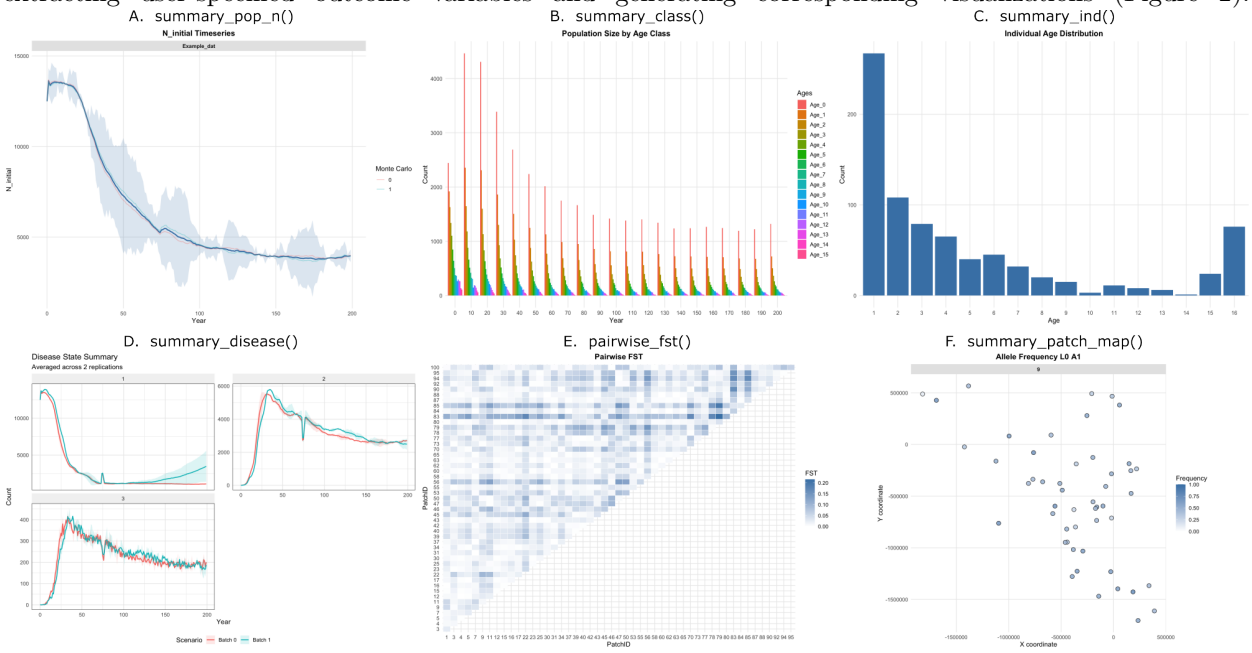


Figure 1: Figure 1: cdmetapop.R package structure.

summary_states(), which analyzes disease state dynamics from summary_popAllTime_DiseaseStates.csv; 6) allele_frequencies_ind(), heterozygosity_ind(), and pairwise_fst_ind(), which summarize individual-level genetic metrics; and 7) summary_patch_map(), which represents various metrics spatially. These functions are designed to facilitate the analysis of CDMetaPOP output and provide users with initial insights into their simulations. These functions are designed to handle both in-memory and file-based workflows, accepting either a dataframe or a file path as input. They can also process outputs recursively from multiple directories (e.g. Monte Carlo replicates or different species), extracting user-specified outcome variables and generating corresponding visualizations (Figure 2).



The package leverages on several domain-standard R packages for efficient data wrangling and transformation (dplyr and tidyr, reshape2), spatial (terra, graph4lg and gdistance) and population genetic (adegenet, poppr) analysis, visualization (ggplot2). An interactive user interface is provided with the use of shiny and shinyBS. Wickham et al. (2024), Wickham and Vaughan (2024), Wickham (2014), Hijmans (2024), Team (2021), Etten (2017), Jombart (2008), Kamvar et al. (2014), Wickham (2016), Chang et al. (2024), Bailey (2015).

The package integrates with the broader R/CRAN ecosystem, including documentation generated with roxygen2 and testing implemented using testthat. Wickham, Danenberg, et al. (2024), Wickham (2024).

Research impact statement

Spatially explicit, individual-based models (IBMs) are a powerful tool for understanding complex ecological and evolutionary processes, as they allow researchers to simulate the behavior and interactions of individual organisms across space and time. CDMetaPOP is a widely used, open source IBM that combines demography and genetics in a spatially explicit framework, making it particularly useful for studying population dynamics, gene flow, selection and adapation, disease spread, and the effects of landscape features on populations. CDMetaPOP has a steep learning curve and is still underutilized in the research community. This package is designed to make CDMetaPOP more accessible to a wider audience of researchers, including those who may not have extensive experience with Python or command-line interfaces. By providing user-friendly tools for input file creation and analysis of CDMetaPOP output in the more commonly used R programming language, the package aims to promote broader adoption of CDMetaPOP across multiple research disciplines.

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AI usage disclosure

Portions of the R package code were developed with assistance from OpenAI Codex. All AI-generated code and text were reviewed, modified, tested, and validated by the authors. The authors made all substantive design, implementation, and scientific decisions and take full responsibility for the software and manuscript content.

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